



Aya Yousef

AI/ML Engineer | Deep Learning & NLP | Full-Stack Development

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SUMMARY

AI/ML-focused software engineer with an MSc (Honors) in Software Engineering from ITMO University and a BSc in Information Technology from Tartous University, Syria. I specialize in applying machine learning to practical, real-world applications, with a strong foundation in modern web and mobile development. My technical foundation spans Python, JavaScript, TypeScript, React, and React Native, complemented by strong fundamentals in object-oriented programming, version control, and software testing. I build user-centric web and mobile applications, with a focus on bringing machine learning capabilities directly into the browser for fast, private, on-device experiences. I'm passionate about writing clean, performant code and applying ML in practical, deployable ways — whether that's a responsive web app, a cross-platform mobile experience, or an ML-powered feature running entirely client-side.

EXPERIENCE

Freelance AI/ML Engineer — Self-employed

Mar 2026 – Present

Remote

- Integrated a browser-based sentiment analysis model, developed through Master's thesis research, into the company's website to automatically classify customer reviews.
- Designed the system to retain only the sentiment classification and a short review summary — not the full review text — improving customer data privacy and reducing server-side processing/storage load.
- Delivered results formally recognized via a signed Act of Implementation (Jun 2026) confirming practical adoption of the research.

PROJECTS

Research on the Application of Machine Learning Methods for Natural Language Processing in the Frontend

Sep 2024 – Jun 2026

Associated with ITMO University

- Built a full-stack multilingual (English, Arabic, Russian) sentiment analysis system running entirely client-side, eliminating server inference.
- Fine-tuned DistilBERT, converted to ONNX with INT8 quantization — 75% smaller with negligible quality loss; achieved 94.91ms latency and 0.72 F1-macro.
- Implemented multi-threaded in-browser inference (React, TypeScript, WebAssembly, Web Workers) with automatic WebGPU/WASM fallback.
- Published the model on Hugging Face and deployed via CI/CD (GitHub Actions) to Yandex Serverless.

Technologies: Python, PyTorch, Hugging Face Transformers, ONNX Runtime Web, React, TypeScript, WebAssembly, Web Workers, WebGPU

My Car

Mar 2023 – Sep 2023

Associated with Tartous University

- Built a Flutter companion app for an in-vehicle sensor device, visualizing live location, status, weather, and malfunction alerts.
- Implemented multi-language support and theme customization.

Technologies: Flutter, Firebase, Front-End Development, Mobile App Development

EDUCATION

MSc (Honors), Web Technologies — ITMO University

Oct 2023 – Jul 2026

Saint Petersburg, Russia

- Graduated with Honors (Red Diploma), GPA 5.0/5.0.
- Thesis: "Research on the Application of Machine Learning Methods for Natural Language Processing in the Frontend" — moving ML models out of the backend and directly into the browser, instead of relying on server-side API calls.
- Worked across NLP & transformer models, model optimization (ONNX conversion, INT8 quantization), in-browser ML inference (WebAssembly, Web Workers), WebGPU/WASM backend fallback systems, cross-platform development (React, React Native), and automated benchmarking/evaluation pipelines.
- Delivered a full-stack multilingual sentiment analysis system (English, Arabic, Russian) running entirely client-side, with the fine-tuned model published on Hugging Face and the application deployed via CI/CD.

BSc, Information Technology — Tartous University

Oct 2018 – Oct 2023

Tartous, Syria

- Graduated with a grade of 82.15%.
- Activities: ICPC (International Collegiate Programming Contest).
- Program covered mathematics, computing, information systems, software engineering, data systems, and intelligent technologies.

SKILLS

AI/ML & NLP: Artificial Intelligence (AI), Machine Learning, Deep Learning, Natural Language Processing (NLP), Sentiment Analysis, Logistic Regression, Transformer Models, PyTorch, TensorFlow, Model Optimization (ONNX, INT8 Quantization), Data Analysis

On-Device / In-Browser ML: WebAssembly (WASM), WebGPU, Web Workers, Client-Side Inference

Full-Stack & Mobile Development: JavaScript, TypeScript, React.js, React Native, Flutter, Node.js, Front-End Development, Back-End Web Development, Full-Stack Development, Cross-platform Development, Responsive Web Design

Programming & CS Fundamentals: Python (Programming Language), Object-Oriented Programming (OOP), Data Structures, Unit Testing, Debugging

Tools & Infrastructure: Git, GitHub, Version Control, Docker, CI/CD (GitHub Actions), Firebase, Databases

Design: UI/UX Design, Web Application Design

CERTIFICATIONS

- Natural Language Processing with Classification and Vector Spaces — DeepLearning.AI (Jul 2026)
- Programming with JavaScript — Meta (Aug 2025)
- Introduction to Mobile Development — Meta (Jul 2025)
- Version Control — Meta (Jul 2025)
- Programming in Python — Meta (Jul 2025)
- Introduction to Back-End Development — Meta (Jul 2025)
- Introduction to Front-End Development — Meta (Jun 2025)
- Get Started with Figma — Coursera (Mar 2023)

LANGUAGES

Arabic (Native) · English (B2) · Russian (B1)